

1. What are transformers?

“Transformers are the neural network architecture that transforms input sequence into output sequence by learning context and tracking relationships between components”

- This are the foundation of modern AI models like BERT,GPT-3 and etc,.
- Transformers process inputs through multiple identical building blocks stacked in layers .
- Each block should contain 2 layers attention layer and feed forward network.
- The transformers consists of two main components encoder stack and decoder stack

Key components :

1. Tokenization &input embeddings
2. Positional encoding
3. Self attention mechanism
4. Multi head attention
5. Feed forward network (MLP layer)
6. Encoder - decoder architecture
7. Normalization and residual connections
8. Output

2.Why were transformers introduced ?

Transformers were introduced to address key limitations of previous sequence models.

Limitation of RNNs/CNNs	Transformer Solution
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Sequential processing - RNNs process words one-by-one, making training slow	Parallel processing - All words processed simultaneously
Long-term dependency problem - RNNs struggle with relationships between distant words	Self-attention - Directly captures relationships regardless of distance
Limited parallelization - CNNs and RNNs can't fully utilize GPUs	Highly parallelizable - Requires significantly less training time

3. Basic architecture of transformers

Transformers are built from two main blocks those are encoder and decoder ,connected through attention mechanisms the original architecture uses 6 encoders layers and 6 decoder layers

Work flow :

Input → Embeddings → Positional Encoding → Encoder Stack (×N) → Decoder Stack (×N) → Softmax → Output

4. Role of attention mechanism

The attention mechanism is the core innovation that makes Transformers work .

Key Function	Description
Focus on relevant parts	Determines relative importance of different input parts

Dynamic weighting	Dynamically weights and combines input elements
Self-attention	Finds relationships between words in a sentence
Position-independent	Understands word relationships regardless of position

Self-attention specifically determines which words in the input sequence are most relevant to each other, guiding the network to focus on those words